

MATI: Mixture Experts w/ Test-Time Aggregation for Imbalanced Regression

Problem: Tabular imbalanced regression

- Existing methods assume **balanced** test distribution
- Real-world test distribution may differ significantly

Key Ideas:

- Decompose training label distribution via **GMM**
- Generate region-specific synthetic data via **SMOBN**
- Train **multiple experts**, each specialized in a region
- At test-time: adjust aggregation weights **without labels**, based on perturbation stability

Baseline model: TabNetRegressor

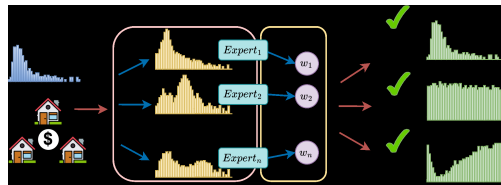


Figure 1(b) from the paper

Method 1 — Region-Aware Mixture Expert Training

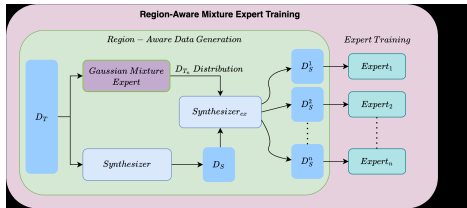


Figure 2 (left): Expert training pipeline

Step 1 — Overall Synthesis (Eq. 1)

$D_S = \text{Synthesizer}(D_T, Y)$ SMOGN-based oversampling

Step 2 — GMM Decomposition (Eq. 2)

$$\text{GMM}(D_T) = \sum_{n=1}^N \pi_n \mathcal{N}(Y | \mu_n, \sigma_n)$$

N determined by AIC (Akaike Information Criterion)

Step 3 — Data Assignment (Eq. 3–4)

$$z_i = \arg \max_n \pi_n \mathcal{N}(y_i | \mu_n, \sigma_n), \quad D_T = \bigcup_{n=1}^N D_{T_n}$$

Step 4 — Per-Expert Synthesis (Eq. 5)

$$D_{S_n} = \text{Synth}_{ex}(D_S, \mu_n - \alpha\sigma_n, \mu_n + \alpha\sigma_n)$$

Oversample component n 's region from the balanced D_S

Step 5 — Expert Training

Each $Expert_n$ trained on D_{S_n} with **MSE loss**

Method 2 — Test-Time Self-Supervised Expert Aggregation

All expert parameters **frozen**; only aggregation weight **w** is learned.

Continuous Prediction Gap Minimization (Eq. 6):

- 1 Generate two perturbed views x_1, x_2 from test sample x via **VIME**
- 2 Forward through each expert $\rightarrow$ logits v_i^1, v_i^2
- 3 Normalize via $\text{softmax}(\mathbf{w})$, compute weighted sums:
 $\hat{y}^{(1)} = \sum_i \sigma(w_i) \cdot v_i^1, \quad \hat{y}^{(2)} = \sum_i \sigma(w_i) \cdot v_i^2$
- 4 Minimize:
$$\min_{\mathbf{w}} S = \frac{1}{|D_t|} \sum_{x \in D_t} (\hat{y}^{(1)} - \hat{y}^{(2)})^2$$

Intuition: Expert proficient in a region $\rightarrow$ stable prediction under perturbation $\rightarrow$ naturally receives larger weight $\rightarrow$ adapts to current test distribution.

Early stop: if any $w_i \leq 0.05$.

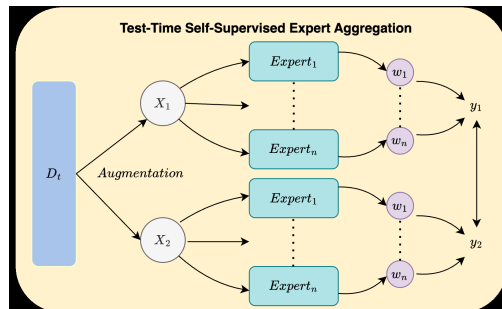


Figure 2 (right): Test-time aggregation

Datasets (tabular)

- Taiwan House (unavailable)
- Abalone — 4,117 samples, 8 features
- Bike Sharing — 17,389 samples, 13 features

Test distributions: balanced / normal / inverse

Code: Not available

Key Results

- Average MAE improvement: **7.1%** over baselines
- Each expert achieves low error on its assigned region
→ GMM-based partitioning is **valid**
- Existing methods: good only on balanced test;
degrade on normal / inverse
- MATI: **consistent** across all three distributions

Ablations: perturbation ratio effect; expert weight distribution

[Tables 1, 2, 3 to be inserted]

- 1 Very small tabular datasets → could have tried **SMOBN** instead of SMOTE for our augmentation.
SMOBN: regression-specific; Gaussian noise for rare regions + SMOTE-style interpolation for moderate regions.
- 2 Expert aggregation weight mechanism is essentially **SADE extended to regression**.
SADE: test-time expert weighting for discrete classes → MATI: continuous targets.
- 3 GMM-based synthesis is likely too **heavy for image data**.
GMM fitting in high-dimensional feature space is inefficient → best suited for tabular.

ImOOD: Rethinking OOD Detection on Imbalanced Data Distribution

Problem: OOD detection under imbalanced ID distribution

Theoretically identifies a **class-aware bias** $\beta(x)$ between balanced and imbalanced OOD detectors. Proposes **training-time loss** ($\mathcal{L}_{\text{ood}} + \mathcal{L}_{\gamma}$) to correct this bias. Applicable to various OOD detectors as a **unified framework**.

Common failures of existing methods:

- **Tail-class ID** → misclassified as OOD
- **OOD samples** → misclassified as head-class ID

→ Head's large decision space & tail's small decision space jointly damage OOD detection.

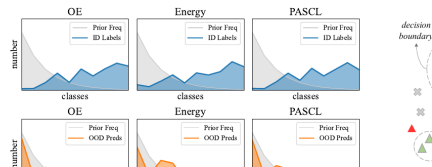


Figure 1(a): Class distribution of wrongly detected ID/OOD samples

Method — Class-aware Bias Between Balanced / Imbalanced OOD Scores

Eq. 1: $P(y|x) \propto P(x|y) \cdot P(y)$

Bayes rule. $P(y)$ is skewed in imbalanced setting.

Eq. 3: $P^{\text{bal}}(y|x) \propto P(y|x) / P(y)$

Class-balanced posterior (prior bias removed).

Eq. 4: $P(i|x) = \sum_y P(y|x) = 1 - P(o|x)$

OOD detection as binary classification.

Thm. 3.2: $P^{\text{bal}}(i|x) = \beta(x) \cdot P(i|x)$

$$\beta(x) = \sum_y \gamma_y(x) \frac{P(y|x, i)}{P(y)}$$

Tail sample $\rightarrow \beta$ large $\rightarrow$ boosts ID score.

OOD predicted as head $\rightarrow \beta$ small $\rightarrow$ suppresses ID score.

Eq. 5: $\sigma(g^{\text{bal}}) = \beta(x) \cdot \sigma(g)$

Sigmoid form of the balanced/imbalanced relation.

Eq. 7 ($\mathcal{L}_{\text{ood}}$):

$$\mathcal{L}_{\text{ood}} = \text{BCE}(g(x) - \Delta(x), t)$$

$$\Delta(x) = \log[\beta + (\beta - 1)e^g]$$

Train g to directly estimate g^{bal} by subtracting bias Δ .
 γ detached. ID: $\Delta \geq 0$, OOD: $\Delta \leq 0$ (clamped).

Eq. 8 ($\mathcal{L}_\gamma$):

$$\mathcal{L}_\gamma = \max\{0, \beta(x) \cdot \sigma(g(x)) - 1\}$$

Probability constraint: $P^{\text{bal}}(i|x) \leq 1$.
 g detached. Only $\gamma_{y;\theta}$ is updated.

At **test time**: use $g(x)$ only.

No β, γ, Δ computation needed.

Experiments

Datasets

- CIFAR10/100-LT (IR=100)
OOD aux: TinyImages 300K
- ImageNet-LT
OOD train: ImageNet-Extra
OOD test: ImageNet-1k-OOD

Architecture

- CIFAR: ResNet18 / ImageNet: ResNet50
- Detector g : shares backbone with classifier f ,
1 additional output node attached
- γ generation: 1 extra `nn.Linear`

Code: github.com/alibaba/imood

Key Results

- CIFAR10-LT: AUROC +1.9%, FPR95 -4.6% (vs. PASCL)
- ImageNet-LT: AUROC +6.7%, FPR95 -13.2%
- Consistent improvement across OE, Energy, Maha detectors
- Outperforms ClassPrior on its own benchmark setting

Ablations

- γ_y : `const` < `class-dep` < `input-dep` (matches theory)
- β, Δ distributions: `tail` $\rightarrow \beta$ large, `OOD` $\rightarrow \beta$ small
- Post-hoc ($\gamma=1$): marginal gain;
training-time regularization is much more effective
(FPR95: 33.39 $\rightarrow$ 30.80 post-hoc vs. 29.95 training)

[Tables 1, 2 to be inserted]

- ① OOD detection is **binary** (ID vs. OOD) performance metric → does not directly tell whether **multi-class** classification improves.
Separate verification needed.
- ② Opposite to MATI: image-based framework → **hard to apply on tabular** data.
Relies on feature extractor (ResNet) + softmax classifier; tabular models have different backbone structures.

PRL: Controllable Long-Tailed Learning via Hypernetwork-Generated Experts

Problem: Long-tailed classification

- Existing methods assume fixed test distribution
→ fail under **distribution shift**
- No mechanism for users to control **head/tail trade-off**

Key Ideas:

- Generate diverse expert classifier heads via **hypernetwork** h_ψ
- Sample from **Dirichlet distribution** to cover all possible distribution scenarios
- Preference vector** α^* at test-time → controllable head/tail trade-off **without retraining**



Figure 1: (a) Traditional, (b) Multi-expert, (c) PRL with hypernetwork + preference control

Theory — Why Diverse Experts Help

Thm. 1 (ERM Limitation):

$$R_{\text{test}}(f) = R_m(f) + \sum_{i=1}^K (\pi_i^{\text{test}} - \pi_i^m) \cdot \mathbb{E}[\ell(f(x), i)]$$

Single-distribution ERM: test risk grows with class prior mismatch.

Def. 2 (ETVD — Environment Total Variation Distance):

$$\delta(E_i, E_j) = \frac{1}{2} \sum_{k=1}^K |\pi_k^i - \pi_k^j|, \quad \Delta(E_1, \dots, E_M) = \max_{i,j} \delta(E_i, E_j)$$

Thm. 2 (Ensemble Bound):

$$R_{\text{test}}(\hat{f}) \leq \frac{1}{N} \sum_{m=1}^N R_m(f_m) + 2M \left(\frac{1}{N} \sum_{m=1}^N \delta(E_m, E_{\text{test}}) + \frac{N-1}{N} \Delta \right)$$

Ensemble of N experts $\rightarrow$ (1) avg. empirical risk + (2) avg. TV distance to test + (3) weighted ETVD.

Diverse experts covering different distributions $\rightarrow$ smaller gap to any test distribution.

Method — Hypernetwork + Stochastic Convex Ensemble + Preference Control

4.2 Diverse Experts

$T=3$ experts share feature extractor ϕ_θ ;
classifier heads generated by hypernetwork:

$$\mathbf{z}_i \sim \text{Dir}(\alpha), \quad \mathbf{w}_i = h_\psi(\mathbf{z}_i)$$

h_ψ : 3-layer MLP with ReLU

Losses (same as SADE):

L_1 : standard CE, L_2 : balanced softmax, L_3 : inverse softmax

4.3 Stochastic Convex Ensemble (SCE)

Promote expert diversity via worst-case loss:

$$\min_{\Theta} \sum_i L_i + \lambda \cdot \log \sum_i \exp(L_i/\lambda)$$

$\lambda \rightarrow 0$: approaches $\min_{\Theta} \max_p \sum p_i L_i$

Second term penalizes any single expert dominating.

4.4 Preference-Controlled Trade-off

At test time, user inputs $\alpha^* = (\alpha_1^*, \alpha_2^*, \alpha_3^*)$:

$$\hat{\mathbf{r}} = \frac{\mathbf{r} \odot \alpha^*}{\mathbf{r}^\top \alpha^*}$$

$$\hat{\mathbf{W}}_i = h_\psi(\hat{\mathbf{r}}) \rightarrow \text{new classifier heads}$$

α^* controls head/tail focus:

- $R=(1.0, 2.7)$: best on many-shot (head)
- $R=(1.9, 1.1)$: best on few-shot (tail)
- $R=(0.5, 2.5)$: best on medium-shot

No retraining needed — just change α^* and forward through h_ψ .

Pareto-optimal set: covers all possible distribution scenarios; each preference maps to a dedicated solution on the Pareto front.

Experiments

Datasets

- CIFAR100-LT (IR=10, 50, 100)
- Places-LT, iNaturalist 2018
- ImageNet-LT

Backbones

- ResNeXt-50 (ImageNet-LT)
- ResNet-32 (CIFAR100-LT)
- ResNet-152 (Places-LT)
- ResNet-50 (iNaturalist)

Code: github.com/DataLab-atom/PRL

Standard LT Recognition (Table 1)

- SOTA on all 4 datasets (uniform test)
- CIFAR100-LT: +0.6–0.8% vs. LSC/BalPoE
- ImageNet-LT: **60.8%** (vs. 60.2 LSC, 59.3 BalPoE)

Distribution-Shift LT (Table 2, 3)

- Forward-LT to Backward-LT: consistently best
- Even under extreme backward dist., maintains performance

Ablations

- w/o hypernetwork → performance drops
- w/o Chebyshev polynomial → performance drops
- Both components essential for distribution adaptation

[Tables 1, 2, 3, 4 to be inserted]

- 1 Extends SADE paradigm: **expert diversity + test-time adaptation + preference control**.
SADE: fixed experts $\rightarrow$ PRL: hypernetwork-generated experts with controllable trade-off.
- 2 Hypernetwork generates only classifier heads $\rightarrow$ **shared backbone**, minimal extra cost.
 h_ψ : 3-layer MLP; classifier head $w_i \in \mathbb{R}^{d \times C}$ only.
- 3 Preference vector semantics are intuitive: α^* directly maps to head/tail emphasis.
Practical value: medical screening (favor tail) vs. routine classification (favor head).
- 4 Classification + softmax-based $\rightarrow$ **not directly applicable to regression** (cf. MATI).